Department of Computer Science & Engineering

5th Semester

S.No.	Code	Course Title	L-T-P	Credits
1	CS1305	Computer Networks	3-0-0	3
2	CS1307	Distributed Computing	3-0-0	3
3	CS13XX	Professional Elective - I	3-0-0	3

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SYLLABI

5th Semester

Department of Computer Science & Engineering

CS1305 3-0-0 (3)

Computer Networks

Pre-Requisites: CS1208

Course Outcomes:

CO-1	**Understand OSI and TCP/IP models.**
CO-2	Implement routing and congestion control algorithms.

Course Articulation Matrix:

	PO-1	**PO-2**	**PSO-1**
CO-1	3	2	-
CO-2	1	-	2

1 - Slightly; 2 - Moderately; 3 - Substantially

Syllabus:

Introduction - network architecture - reference models. Network layer - routing algorithms - congestion control. Transport layer - TCP - BSD sockets.

Learning Resources:

Text Books:

1. Larry L Peterson, Bruce S Davie, "Computer Networks", 5th Edition, Elsevier, 2012.

2. Andrew S. Tanenbaum, David J Wetherall, "Computer Networks", 5th Edition, Pearson, 2010.

Reference Books:

1. W. Richard Stevens, TCP/IP Illustrated, Volume 1, Addison-Wesley, 2011.

CS1309

0-1-2 (2)

Computer Networks Laboratory

Pre-Requisites: None

Syllabus:

Socket programming; simulation of routing protocols; packet capture and analysis.

Learning Resources:

Text Books:

1. Kevin R. Fall, W. Richard Stevens, TCP/IP Illustrated, Volume 1, 2011.

Professional Electives

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CS1325 3-0-0 (3)

Artificial Intelligence

Pre-Requisites: None

Syllabus:

Problem solving by search; knowledge representation; planning; learning.

Learning Resources:

Text Books:

1. Stuart Russell, Peter Norvig, "Artificial Intelligence: A Modern Approach", 4th Edition, Pearson, 2020.